Irfan Ali

SENIOR GIS DEVELOPER

Transferable Iqama

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PROFILE

Senior GIS Developer with 8 years in software engineering, building enterprise Esri GIS solutions for Saudi government and infrastructure clients including **ESNAD**, **Ministry of Industry and Mineral Resources**, **Ministry of Hajj (KIDANA)**, **Ministry of Culture**, **SDAIA**, **NIC**, and **Riyadh Airports**. Builds custom web GIS portals, spatial automation, and enterprise geodatabases, and runs ArcGIS Enterprise from service publishing to high availability and disaster recovery. Works directly with stakeholders from requirements through delivery and production support, and uses **AI tools** every day to design, code, test, and document faster.

SKILLS

Esri / ArcGIS	**ArcGIS Enterprise, ArcGIS Portal, ArcGIS Server**, **Enterprise Geodatabase**, **ArcGIS Maps SDK for JavaScript**, **ArcGIS REST API**, **Experience Builder**, **ArcGIS Dashboards**, ArcGIS Pro, Survey123
Web GIS	Web maps, 2D/3D MapViews, feature and map services, geometry creation and editing, **spatial analysis**, coordinate reference systems, print templates, GIS reports
GIS Admin	Service publishing and security, users/roles/permissions, **multi-machine Server sites, Portal HA**, web adaptor and load balancer, geodatabase replication, **DR and failover testing**, SSL, Active Directory / SSO
Automation	**Python, ArcPy**, geoprocessing, spatial automation and auto deduction engines
Databases	**Microsoft SQL Server** (spatial SQL, T-SQL), **PostgreSQL**, database design, Redis, Elasticsearch
Development	C#, .NET Core, ASP.NET Core Web API, Angular, React, TypeScript, JavaScript, REST APIs, web services
AI Tools	**Claude, Claude Code, OpenAI Codex, ChatGPT, GitHub Copilot**, **MCP connectors**, prompt engineering, AI-assisted coding, code review, testing, and documentation
Security & DevOps	OAuth 2.0, OIDC, JWT, RBAC/ABAC, Azure, Docker, Azure DevOps, Git, CI/CD

WORK EXPERIENCE

Senior GIS Developer, ESNAD - Saudi Mining Services Company

July 2025 - Present

Riyadh, Saudi Arabia

- Designed the GIS solution and data-flow architecture for digital mining license issuance under the **Ministry of Industry and Mineral Resources**, covering license boundaries, mining sites, and approval workflows.
- Built a custom GIS portal with the **ArcGIS Maps SDK for JavaScript** (MapViews, layers, geometry creation and editing) and apps in **Experience Builder** and **ArcGIS Dashboards**.
- Built a **spatial auto deduction engine** as a REST API that automatically adjusts license geometries against business rules and existing boundaries.
- Administered **ArcGIS Enterprise**: published and secured map and feature services, managed **Portal users, roles, and permissions**, and configured **SSL and Active Directory / SSO**.
- Designed the on-premises **high availability and disaster recovery** setup: multi-machine ArcGIS Server site, Portal primary/standby, web adaptor behind a load balancer, **geodatabase replication**, and failover testing.
- Integrated the GIS platform with other **government systems via REST APIs** for licensing approvals, and exposed spatial endpoints for other departments to consume.
- Built print templates and geoprocessing automation with **Python and ArcPy**; created an AI search that turns plain-language questions into SQL over GIS data.
- Use **Claude Code, Codex, and GitHub Copilot** daily for coding, refactoring, and reviews, and connect AI assistants to internal tools and data through **MCP connectors** to speed up GIS delivery.
- Wrote solution designs and technical specs, coordinated with infrastructure, data, and security teams, and presented solutions to client stakeholders.

Senior GIS Developer, Geotech Overseas

Oct 2023 - June 2025

Riyadh, Saudi Arabia

- Delivered **KIDANA Permits** for the **Ministry of Hajj**, a GIS-based land allocation system where holy-site plots are assigned to companies through approval workflows.
- Built GIS maps and layers for **Riyadh Airports**, and delivered GIS work for **SDAIA** and the **Ministry of Culture**.
- Architected a **multi-tenant Land Information System** with Esri web mapping, parcel and valuation tools, enterprise geodatabase design, and **ArcGIS Dashboards** for government clients.
- Built the **Caller Location Service** for the **National Information Center (NIC)**, cutting location lookups from 4 seconds to 0.44 ms with RabbitMQ and Memcached.
- Published and secured services on **ArcGIS Server and Portal**, set up HA and DR for on-premises deployments, and collected field data with **Survey123**.
- Led requirement sessions with clients, prepared technical documentation, and coordinated vendors and internal teams through delivery.

Senior Software Engineer (Remote), Intellyct

March 2023 - Sep 2023

Vaughan, Canada

- Designed and implemented **RESTful APIs** using **.NET Core** and **microservices** architecture, breaking down monolithic apps.
- Integrated **Redis**, Azure Service Bus, Kafka, and Azure Cosmos DB to enhance performance, enable real-time messaging, and ensure high availability in distributed environments.
- Implemented **health checks** and monitoring for proactive issue detection and high availability.
- **Docker**-containerized apps for efficient scaling and deployment.

Software Engineer / Senior Software Engineer, Nasfia (Pvt) Ltd.

Sep 2021 - Feb 2023

Karachi, Pakistan

- Led a team to **design, develop, and deploy** company products on **Azure App Service** and **VMs**, improving scalability and reducing operational costs.
- Migrated systems from Azure virtual servers to App Service, integrating Azure Functions, **CI/CD pipelines**, **Blob Storage**, **Logic Apps**, and **Azure SQL Managed Instance** for cost savings and reliability.
- Optimized application performance through query tuning, database indexing, and code refactoring, reducing load times by **20%**.
- Designed and integrated multiple CRM projects using **ASP.NET, SQL Server**, Telerik, HTML5, CSS3, Bootstrap, and JavaScript.
- Reduced support tickets through effective bug fixes in CRM.
- Applied **TDD** to develop sustainable, maintainable code with business value.

Associate Software Engineer, Ficiali

Jan 2018 - Aug 2021

Karachi, Pakistan

- Developed and maintained features across web applications, fixing bugs and implementing enhancements.
- Built and integrated **REST APIs and SOAP APIs**, ensuring end-to-end functionality with front-end integration.
- Wrote **automated scripts** and procedures for software maintenance and background tasks.
- Supported the team by replicating and resolving client-reported issues in the staging environment.

EDUCATION

Bachelor of Science in Computer Science, FAST-NUCES

Aug. 2016 - Sep. 2020

National University of Computer and Emerging Sciences, Karachi, Pakistan

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CERTIFICATIONS

- ArcGIS Python Scripting - LinkedIn Learning, Jun 2026
- System Design: Designing Scalable Systems - LinkedIn Learning, Mar 2026
- Advanced Software Architecture - LinkedIn Learning, Apr-Jun 2025

SOFT SKILLS

Client-facing communication and presentations, stakeholder management, technical ownership, problem solving, explaining technical work to non-technical audiences.